# NEW APPROACH TO MODEL THE MECHANICAL BEHAVIOUR OF THE WORK MATERIAL IN MACHINING

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**Keywords:** Numerical model, mechanical behavior, experimental tests, machining

## 1. INTRODUCTION

A significant number of numerical methods, especially the Finite Element, are being developed and used to simulate the machining operations. However, their efficiency to predict the machining performance depends on the accuracy of material models describing the mechanical behavior. The aim of the work is hence to characterize the mechanical behavior of OFHC copper in machining through experimental tests reproducing identical conditions as those in machining and to identify the constants of constitutive model through numerical simulation.

## 2. BACKGROUND

To describe the mechanical behaviour of a work material during the cutting process, Astakhov (1998) explained that the work material undergoes large plastic deformation and the chip formation is due to ductile damage. Abushawashi (1998) used Johnson-Cook plasticity law (Johnson-Cook, 1983) and Johnson-Cook damage law (Johnson-Cook, 1985) as model to describe the mechanical behaviour for lead samples.
Nevertheless, such approach has never been conducted in work materials which is very ductile as OFHC copper.

## 3. SCIENTIFIC APPROACH

### 3.1 Main concepts

The main concept of the work is to split up the numerical model of the work material during the cutting process into two parts: the plasticity constitutive model and the damage constitutive model. The plasticity constitutive model chosen is the Johnson-Cook plasticity law which uses the total strain, strain rate and temperature as parameters. On the other hand, the damage constitutive model chosen is the Johnson-Cook damage law which uses the stress state (stress triaxiality), strain rate and temperature as parameters.

### 3.2 Tools and Experimental techniques

Two compression test machines will be used to conduct the experiment. The Gleeble compression test machine and the conventional compression test are used to determine the constants of the plasticity law and damage law respectively.
Concerning the numerical simulation, Abaqus CAE will be used to compare with the results acquired from compression tests.

### 3.3 Schedule

The first phase of the project, which will last for two months, is to analyze the results for the compression tests carried out using the Gleeble machine in Angers. The second phase of the project which has a duration of 2 ½ months, consists of preparing experimental test and analyzing the results. Last but not least, the third phase of the work which will last for a week, aims to simulate numerically the cutting process of OFHC copper.

## 4. CONCLUSION

The main objective of the work is to provide the OFHC copper model for a PhD thesis to simulate more precisely the cutting process of such material. The work is divided into two parts which comprises the plasticity model and the damage model of the work material during the cutting process. Since there is not much research working on OFHC copper, this work might be useful for further development.

## 5. REFERENCES

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